# INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer and Communication Engineering

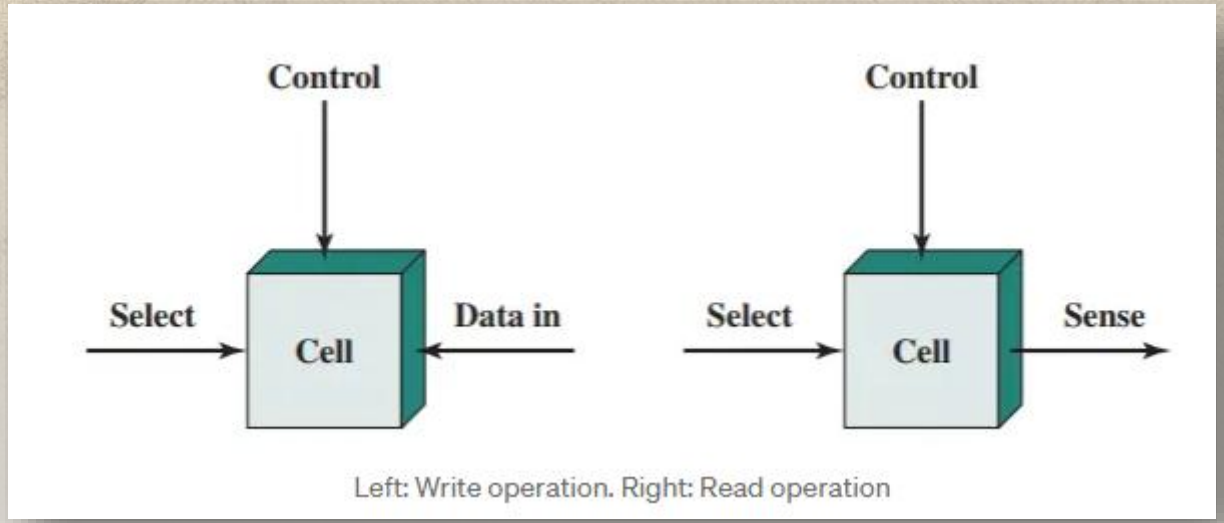
Course Code: CCE-3509
Course Title: Computer Architecture

Abu Noman Siddike
Adjunct Faculty
Dept. of CCE, IIUC

# INTERNAL MEMORY

## Memory Organization

- Refers to the structure and hierarchy of how memory is arranged in a computer system.
- Organized into **levels**: registers → cache → main memory (RAM) → secondary storage.
- Goal: balance **speed, cost, and capacity**.
- Example: Registers (fast, small, expensive) vs. Hard drives (slow, large, cheap).



Left: Write operation. Right: Read operation

# ARM CACHE ORGANIZATION

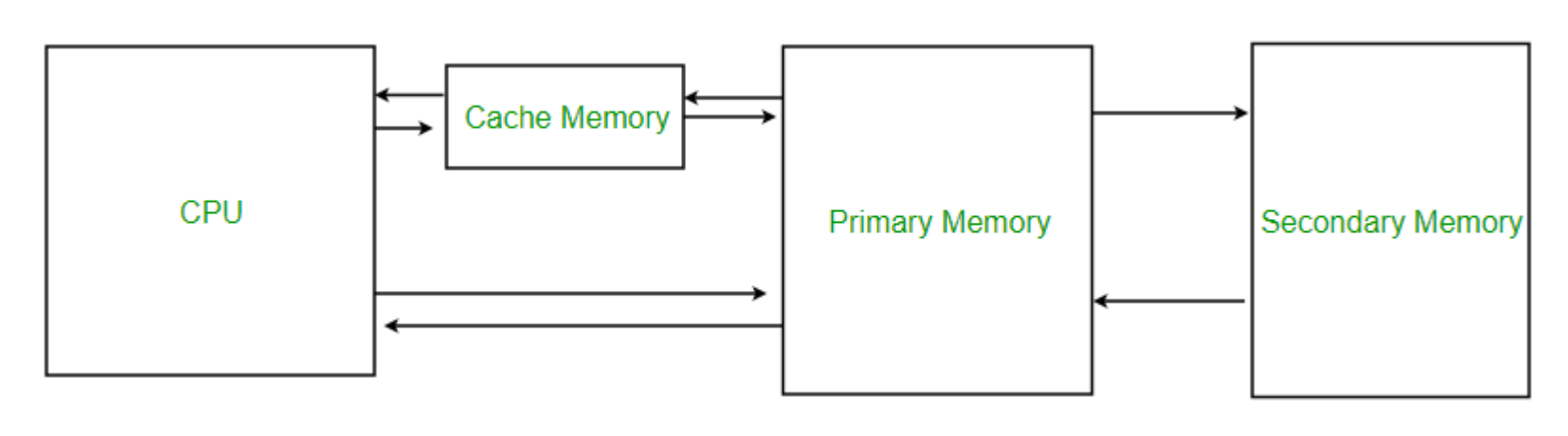
- ARM processors use **multi-level caches** (L1, L2, sometimes L3).
- **L1 cache**: split into instruction cache (I-cache) and data cache (D-cache).
- **Write policies**: Write-through (immediate update to memory) and Write-back (update later).
- ARM designs often optimize for **low power consumption**, essential in mobile devices.
- Example: ARM Cortex-A processors use separate 32 KB L1 instruction & data caches.

# CACHE

A **small, fast memory** between CPU and RAM.
Stores frequently accessed instructions/data to reduce average memory access time.

Mapping techniques:

- Direct mapping (each block maps to one cache line).
- Associative mapping (any block to any line).
- Set-associative mapping (group-based compromise).
- **Example:** If the CPU requests memory location 200, cache may already have it stored, avoiding slower RAM access.



# VIRTUAL MEMORY

Virtual memory is a memory management technique used by operating systems to give the appearance of a large, continuous block of memory to applications, even if the physical memory (RAM) is limited and not necessarily allocated in contiguous manner.

- ✓ creates the illusion of a large memory, even if the actual memory (RAM) is small.
- ✓ uses both RAM and disk storage to manage memory
- ✓ Only the needed parts are loaded.
- ✓ When a program runs, it uses virtual addresses (not real memory locations).
- ✓ converts these virtual addresses into physical addresses (actual locations in RAM) while the program runs.

# TYPES OF VIRTUAL MEMORY

**Paging:** Paging divides memory into small fixed-size blocks called pages. When the computer runs out of RAM, pages that aren't currently in use are moved to the hard drive, into an area called a swap file. Here,

- The swap file acts as an extension of RAM.
- When a page is needed again, it is swapped back into RAM, a process known as page swapping.
- This ensures that the operating system (OS) and applications have enough memory to run.

**Segmentation:** Segmentation divides virtual memory into segments of different sizes. Segments that aren't currently needed can be moved to the hard drive. Here,

- The system uses a segment table to keep track of each segment's status, including whether it's in memory, if it's been modified and its physical address.
- Segments are mapped into a process's address space only when needed.

# Virtual Memory vs Physical Memory

| Feature | **Virtual Memory** | **Physical Memory (RAM)** |
|---|---|---|
| **Definition** | An abstraction that extends the available memory by using disk storage | The actual hardware (RAM) that stores data and instructions currently being used by the CPU |
| **Location** | On the hard drive or SSD | On the computer's motherboard |
| **Speed** | Slower (due to disk I/O operations) | Faster (accessed directly by the CPU) |
| **Capacity** | Larger, limited by disk space | Smaller, limited by the amount of RAM installed |
| **Cost** | Lower (cost of additional disk storage) | Higher (cost of RAM modules) |
| **Data Access** | Indirect (via paging and swapping) | Direct (CPU can access data directly) |
| **Volatility** | Non-volatile (data persists on disk) | Volatile (data is lost when power is off) |

# CHANNEL

**What Is a Channel?**
A **channel** is a dedicated hardware or logical unit that controls and manages data movement between memory and I/O devices. It acts as a **traffic controller**, coordinating communication without burdening the CPU.

| Type | Description |
|---|---|
| **I/O Channel** | Manages communication between CPU and I/O devices. Often includes its own processor. |
| **Memory Channel** | Handles data flow between memory modules and CPU or other subsystems. |
| **DMA Channel** | Direct Memory Access — allows devices to transfer data to/from memory without CPU intervention. |

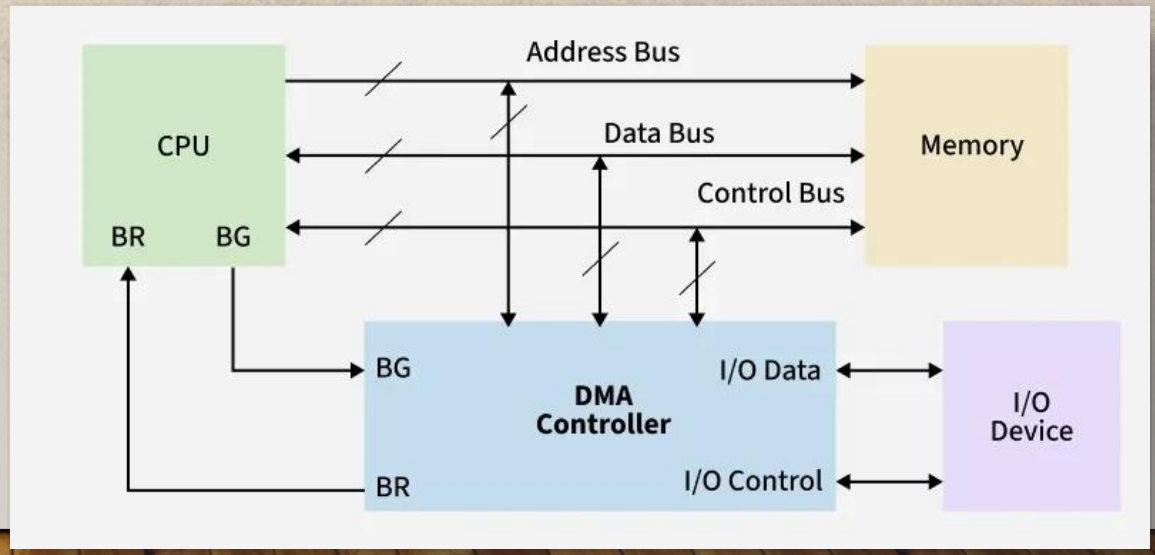
# DIRECT MEMORY ACCESS(DMA)

DMA Controller is a type of control unit that works as an interface for the data bus and the I/O Devices. DMA Controller has the work of transferring the data without the intervention of the processors, processors can control the data transfer.

The DMA controller registers have three registers as follows.

- **Address register:** It contains the address to specify the desired location in memory.
- **Word count register:** It contains the number of words to be transferred.
- **Control register:** It specifies the transfer mode.

.



# DMA

- **The Problem:** Without DMA, the CPU (**Programmed I/O**) must read every byte of data from a device (like a hard drive) and then write it to memory. This wastes precious CPU cycles.
- **The Solution:**
  - A **DMA Controller** is a specialized hardware unit.
  - The CPU sets up a DMA transfer by telling the DMA controller: the *source* address (e.g., device buffer), the *destination* address in memory, and the *amount* of data.
  - The DMA controller then performs the transfer **directly between the device and main memory**, without involving the CPU.
  - Once the transfer is complete, the DMA controller sends an **interrupt** to the CPU to let it know the task is done.
- **Interrupts:** A signal sent to the CPU that alerts it to an event that needs its attention. The CPU **suspends** its current program, saves its state, and executes an **Interrupt Service Routine (ISR)**—a small program to handle the event (e.g., a keypress, packet arrival, DMA completion). After the ISR, it resumes the original program.

# TYPES OF DIRECT MEMORY ACCESS (DMA)

I. **Single-Ended DMA:** In this type, the DMA controller is connected only to one device (usually either the memory or the I/O device), and it directly controls data transfer.

II. **Dual-Ended DMA:** The DMA controller is connected to both the source and the destination, typically memory and an I/O device.

III. **Arbitrated-Ended DMA:** In systems with multiple DMA devices or masters, arbitration is needed to decide which device gets control of the bus. It is more advanced than Dual-Ended DMA.

IV. **Interleaved DMA:** Interleaved DMA are those DMA that read from one memory address and write from another memory address.

# PROS. AND CONS.

**Advantages of DMA Controller**

i. DMA speeds up memory operations and data transfers by allowing peripherals to communicate directly with memory, bypassing the CPU.

ii. During data transfer, the CPU is not involved, which significantly reduces its workload.

iii. Operates efficiently, requiring very few clock cycles to complete data transfers.

iv. It distributes workload very appropriately.

v. By offloading data movement tasks from the CPU, DMA distributes the overall system workload more effectively

**Disadvantages of DMA Controller**

i. Costly operation due to the need for additional hardware and control logic.

ii. Suffers from Cache-Coherence Problems.

iii. It increases the overall cost of the system.

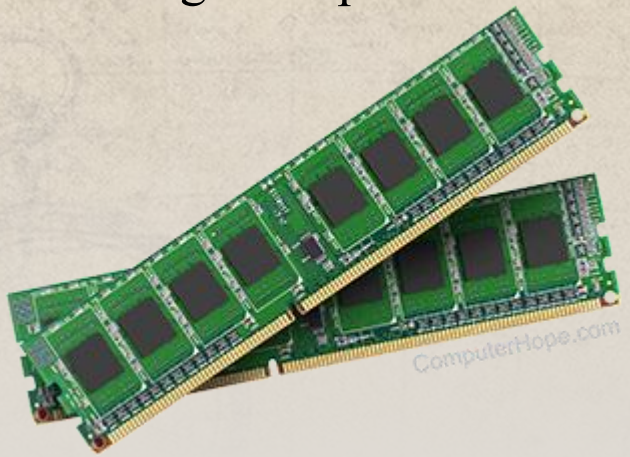
iv. It increases the complexity of the software.

# RAM

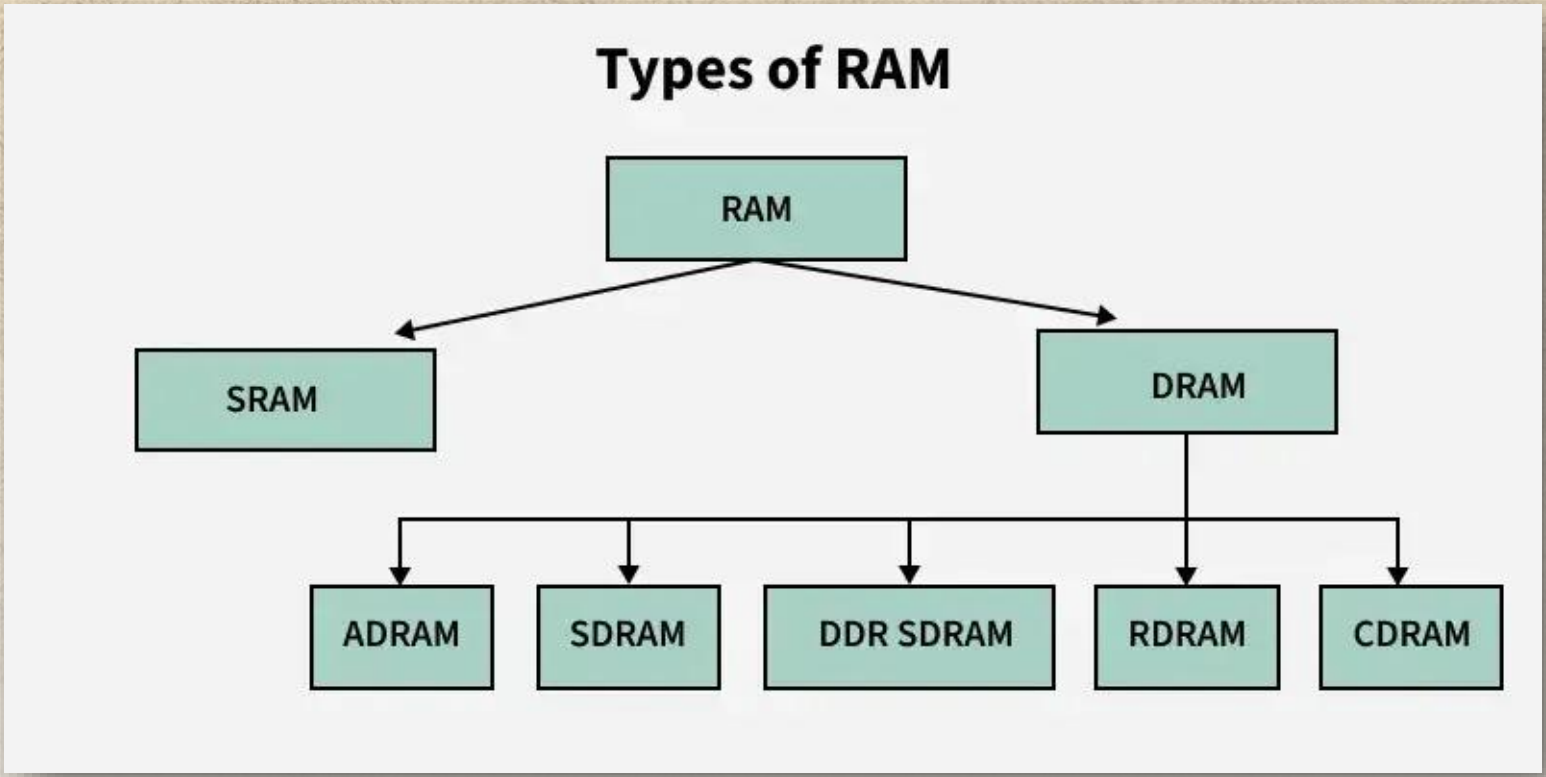
**What is RAM?**

Random Access Memory, is a type of computer memory that allows data to be read and written randomly, meaning that the computer can access any location in the memory directly rather than having to read the data in a specific order.

Fig: Computer RAM



# TYPES OF RAM

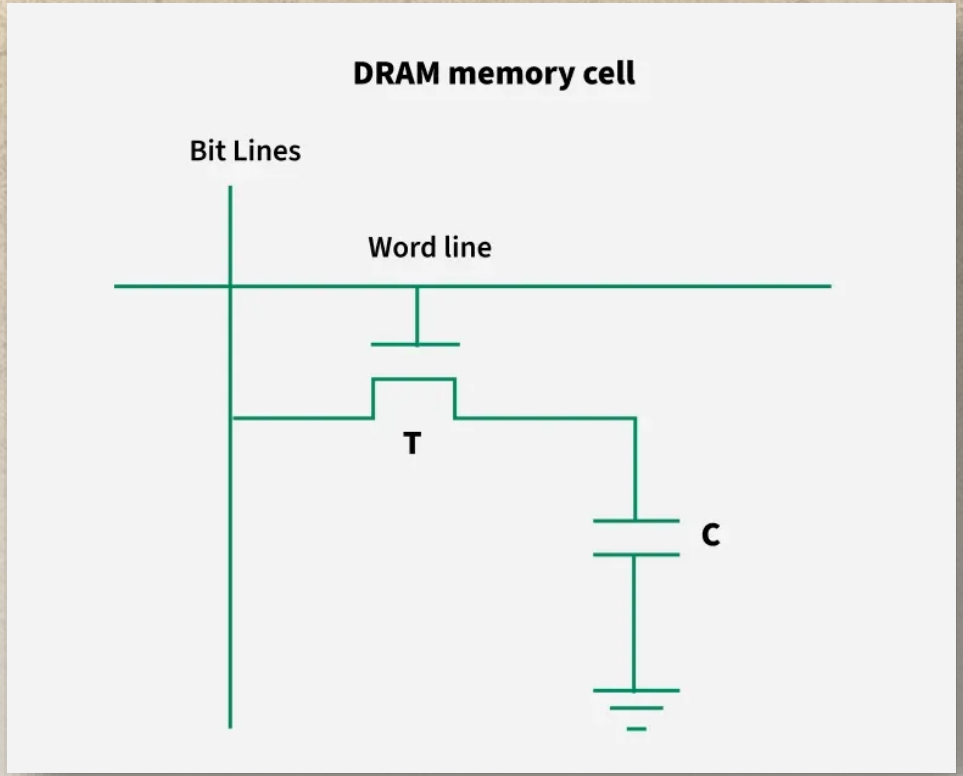


# SRAM VS DRAM

| Aspect | SRAM (Static RAM) | DRAM (Dynamic RAM) |
|---|---|---|
| **Technology Used** | Built using flip-flops (typically 6 transistors per bit); stable and fast | Built using capacitors and a single transistor per bit; simpler but needs refreshing |
| **Refresh Requirement** | No refresh needed — data remains as long as power is supplied | Requires periodic refresh due to capacitor leakage — adds complexity and latency |
| **Speed** | Very fast access times (~1 ns); ideal for performance-critical tasks | Slower access times (~50–70 ns); suitable for bulk storage |
| **Cost** | Expensive due to complex circuitry and larger cell size | Cheaper to manufacture; higher density per chip |
| **Power Consumption** | Consumes more power when active; idle power is low | Lower active power, but refresh cycles add to overall consumption |
| **Typical Usage** | Used in cache memory (L1, L2, L3) close to CPU for rapid access | Used in main memory (RAM modules) for general-purpose data storage |
| **Size Range** | Small — typically 1MB to 16MB | Large — typically 4GB to 16GB or more |
| **Location in System** | Integrated into processor or between CPU and main memory | Located on the motherboard as part of system RAM |
| **Volatility** | Volatile — loses data when power is off | Volatile — also loses data when power is off |

# FEATURES OF DRAM

- ✓ DRAM has to have its contents refreshed constantly to maintain data.
- ✓ It is cheaper compared to other forms of memory such as SRAM.
- ✓ Slower than SRAM.
- ✓ Power consumption in the case of DRAM is less than the other memory type but always greater in comparison to the case of SRAM.
- ✓ DRAM is physically smaller in comparison to other alternatives.
- ✓ It has short data lifetime.



# TYPES OF DRAM

There are mainly 5 types of DRAM.

i. **Asynchronous DRAM (ADRAM):** A specialized memory controller circuit generates the necessary control signals to control the timing. The CPU must take into account the delay in the response of the memory.

ii. **Synchronous DRAM (SDRAM):** These RAM chips' access speed is directly synchronized with the CPU's clock. For this, the memory chips remain ready for operation when the CPU expects them to be ready.

iii. **Double-Data-Rate SDRAM (DDR SDRAM):** This faster version of SDRAM performs its operations on both edges of the clock signal; whereas a standard SDRAM performs its operations on the rising edge of the clock signal.

iv. **Rambus DRAM (RDRAM):** The RDRAM provides a very high data transfer rate over a narrow CPU-memory bus.

v. **Cache DRAM (CDRAM):** This memory is a special type of DRAM memory with an on-chip cache memory (SRAM) that acts as a high-speed buffer for the main DRAM.

# PROS. AND CONS.

- **Advantage of DRAM**
- DRAM memory can be deleted and refreshed while running the program.
- It is Cheaper as compare to SRAM.
- It has higher storage capacity. Hence it is used to create larger RAM space system.
- It is simple in structure than SRAM.

- **Disadvantage of DRAM**
- It is comparatively slower than SRAM. Hence it takes more time for accessing data or information.
- It loses data when power is OFF.
- It has higher power consumption compare to SRAM

# EXTERNAL MEMORY

**A magnetic Disk** is a type of secondary memory that is a flat disc covered with a magnetic coating to hold information. It is used to store various programs and files. The polarized information in one direction is represented by 1, and vice versa. The direction is indicated by 0.

- ✓ These are economical memory
- ✓ Easy and direct access to data is possible.
- ✓ It can store large amounts of data.
- ✓ It has a better data transfer rate than magnetic tapes.
- ✓ It has less prone to corruption of data as compared to tapes.



# RAID (REDUNDANT ARRAYS OF INDEPENDENT DISKS)

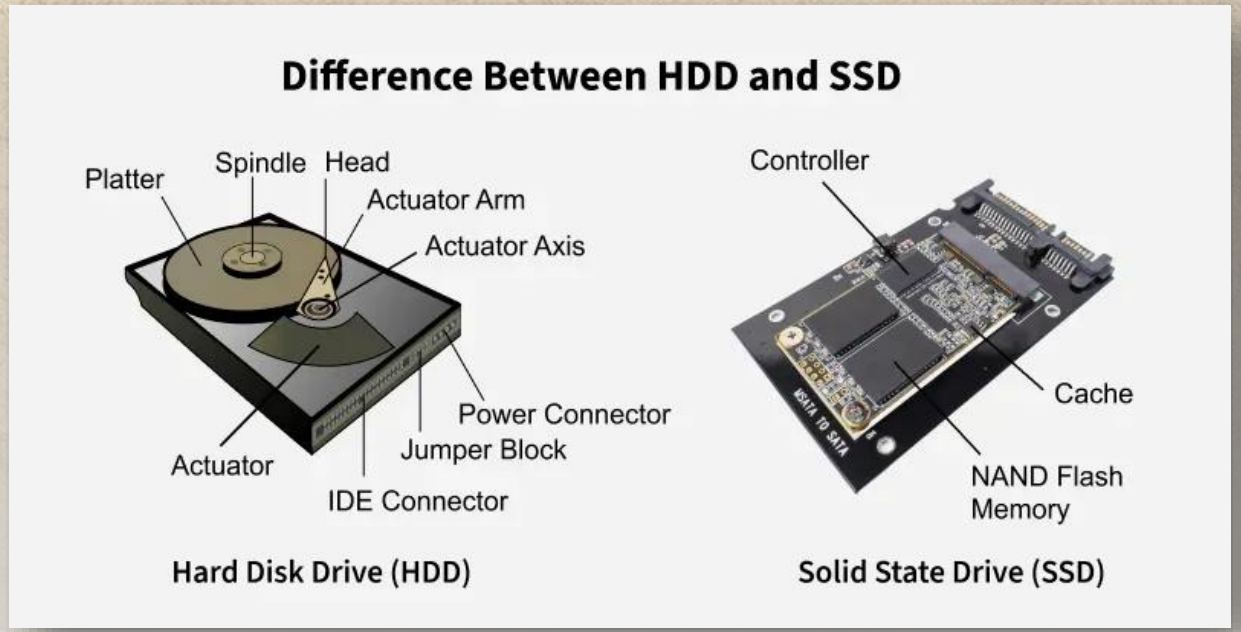
RAID is a technique that combines multiple hard drives or SSDs into a single system to improve performance, data safety or both. If one drive fails, data can still be recovered from the others.

- The data is split into small blocks (like dividing a file into chunks). These blocks are written across multiple drives in parallel.
- **Mirroring (RAID 1):** Exact copy of data is kept on another drive.
- **Parity (RAID 5, RAID 6):** A calculated value (parity block) is stored to allow data recovery in case of failure.
- **Fault Tolerance**: If one drive fails, RAID uses the redundant data (mirror or parity) to reconstruct missing data.

# SOLID STATE DRIVE (SSD)

A Solid-State Drive (SSD) is a non-volatile storage device that stores data without using any moving parts, unlike traditional Hard Disk Drives (HDDs), which have spinning disks and mechanical read/write heads.

- Faster startup time
- Random Access times
- Data Tranfer rate (100-600Mb/s)
- Zero noise due to no mechanical parts
- Reliability
- Secure



# HDD VS SDD

| Parameter | HDD | SSD |
|---|---|---|
| **Long term storage** | HDDs are more reliable for long-term storage. | less reliable for long-term storage due to data leaks that can occur if kept unpowered for more than a year. |
| **Access speed** | The data accessing speed is slower as compared to SSD. | The data accessing speed is much higher as compared to HDD. |
| **Performance** | The performance suffers because of fragmentation. | The performance does not suffer because of fragmentation. |
| **Suitable for** | •Extensive storage<br>•Long-term storage | •Fast data retrieval<br>•Laptop or desktop because of low power consumption and size. |

# OPTICAL MEMORY

Optical memory is a form of electronic storage that uses a laser beam to store and retrieve data. It was developed by Philips and Sony and released in 1982 in the fourth generation of computers. Example: CD, DVD

- Portability
- Durability
- Storage Capacity
- Use case



# MAGNETIC TAPE

Magnetic tape is a medium for magnetic storage made of a thin, magnetizable coating on a long, narrow strip of plastic film.

The width of the ribbon varies from 4mm to 1 Inch and it has storage capacity 100 MB to 200 GB.

**Advantages :**

1. These are inexpensive, i.e., low cost memories.
2. It provides backup or archival storage.
3. It can be used for large files.
4. It can be used for copying from disk files.
5. It is a reusable memory.
6. It is compact and easy to store on racks.



# I/O MODULES

I/O module is the key element for computer system. I/O modules control the peripheral devices. Thus I/O module is required which has two major functions.

- Interface to the processor and memory via the system bus or central switch.
- Interface to one or two peripheral devices by tailored data links

The five major functions of I/O module :

i. Control and timing

ii. Processor communication

iii. Device communication

iv. Data buffering

v. Error Detection

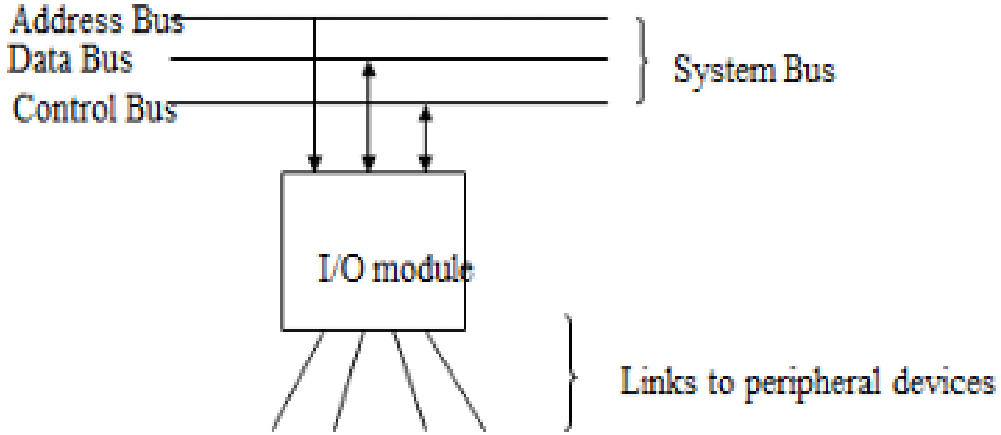



Fig. Generic model of an I/O module

# EXTERNAL DEVICES

**External devices** provide the means of exchanging data between the external environment and the computer.  An external device is attached to computer by a link to an I/O module.

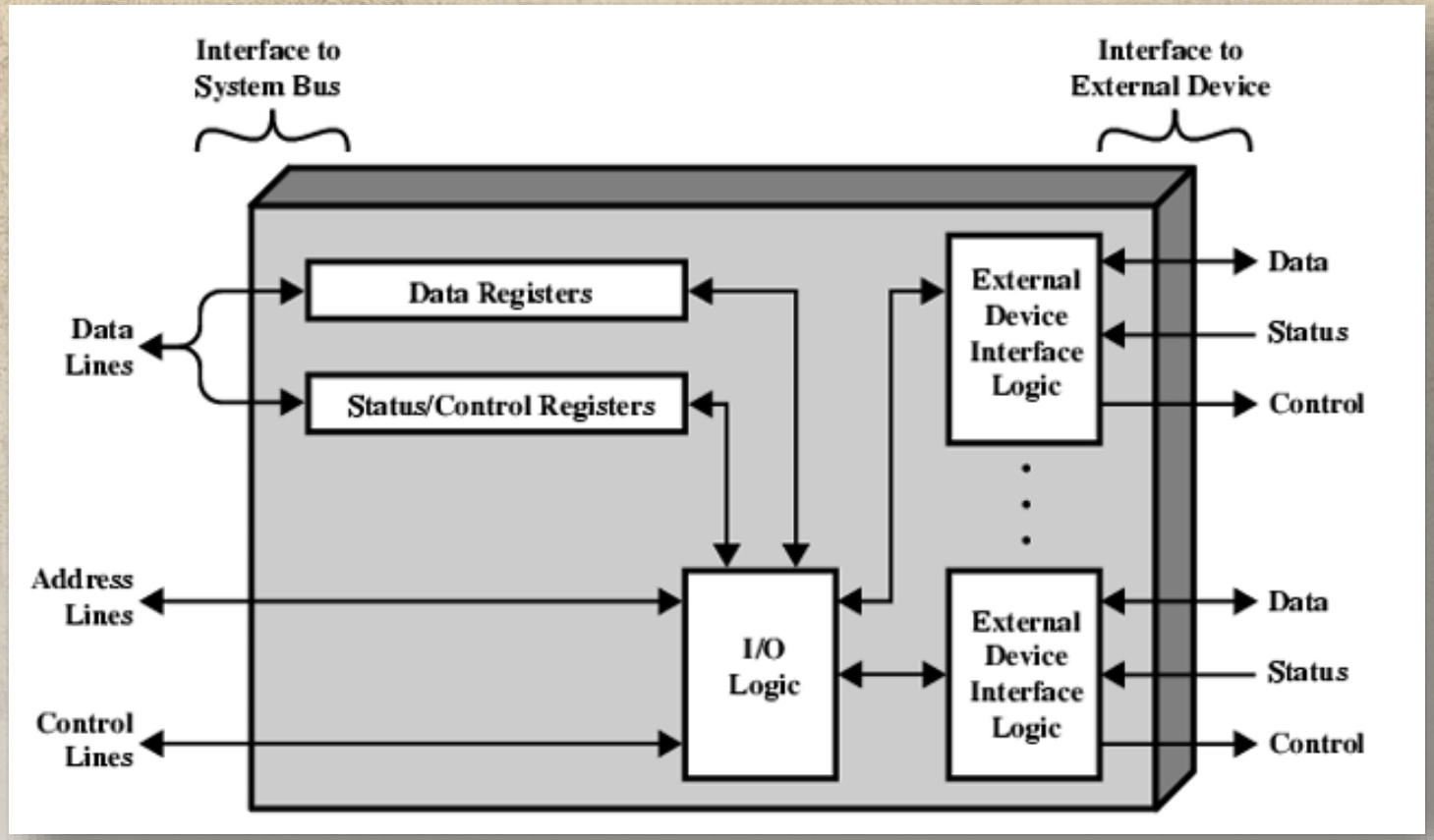
External device is referred to as peripheral device or peripheral. It is classified in three categories:

i. Human-readable: suitable for communicating with computer user. E.g. Screen, Printer, Keyboard etc.

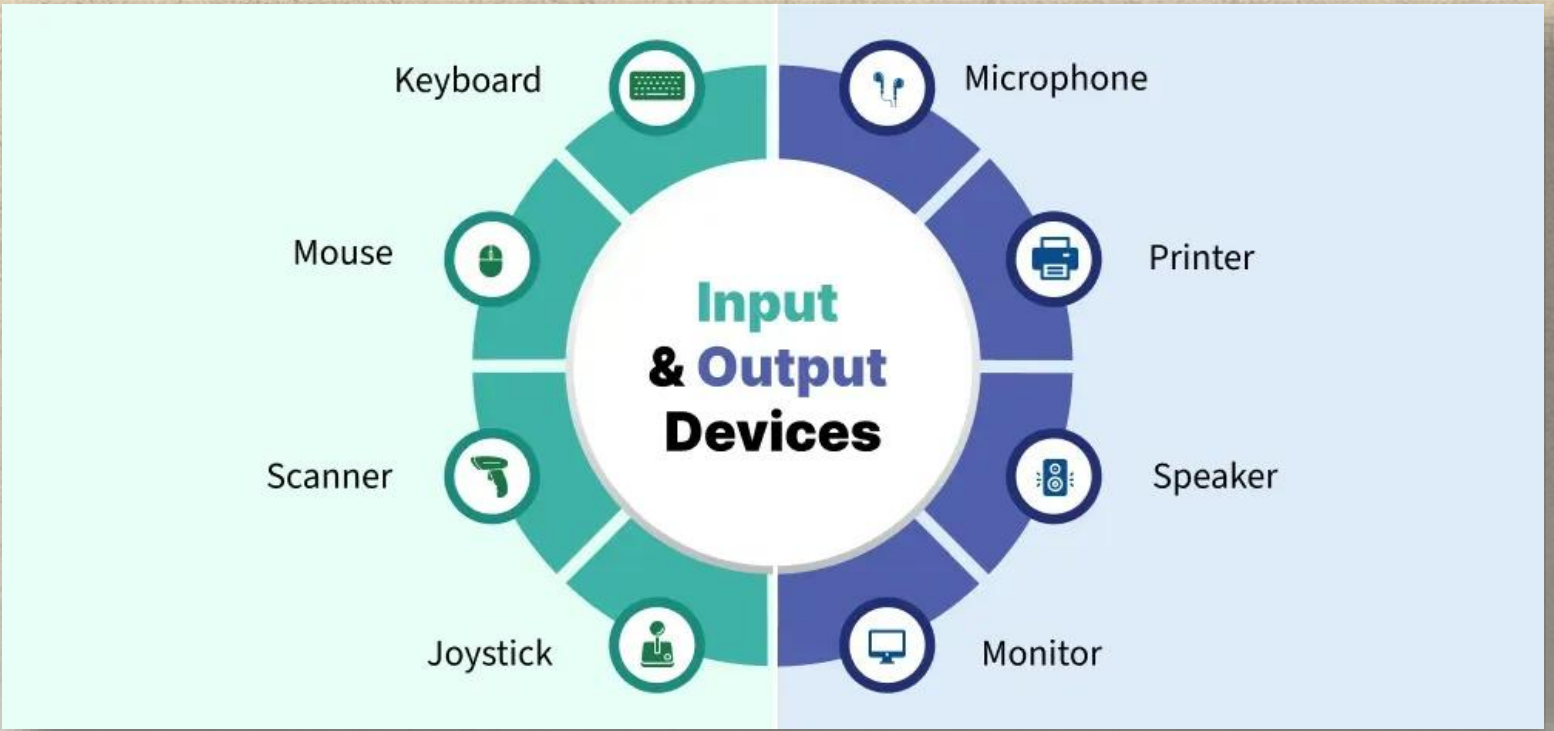
ii. Machine-readable: suitable for communicating with equipment. E.g. magnetic disk, tape, sensor, actuator etc.

iii. Communicational: suitable for communicating with remote devices. E.g. Modem, NIC etc.

# I/O MODULE STRUCTURE



# I/O DEVICES


Keyboard
Mouse
Scanner
Joystick
Input
& Output
Devices
Microphone
Printer
Speaker
Monitor

# I/O TECHNIQUES

Three techniques are possible for I/O operations.

i. Programmed I/O

ii. Interrupt driven I/O

iii. Direct Memory Access (DMA)

# PROGRAMMED I/O

In programmed I/O, the processor executes a program that gives it direct control of the I/O operation, including sensing I/O device status, sending a read or write command, and transferring the data.

**I/O commands:** Four types of I/O command is received:

- **Control** -It activates peripherals to perform required action. eg. Rewind, forward, pause in an magnetic disk etc.
- **Test** - Checks the status conditions associated with an I/O module and its peripherals. Processor checks for the available peripherals. It also checks the most recent operation is completed or not..
- **Read** - I/O module reads data from peripherals and place it into an internal buffer. The processor then obtains data by requesting I/O module to place it on data bus.
- **Write** - I/O module takes data from data bus and transmits the data to peripheral.

# PROGRAMMED I/O

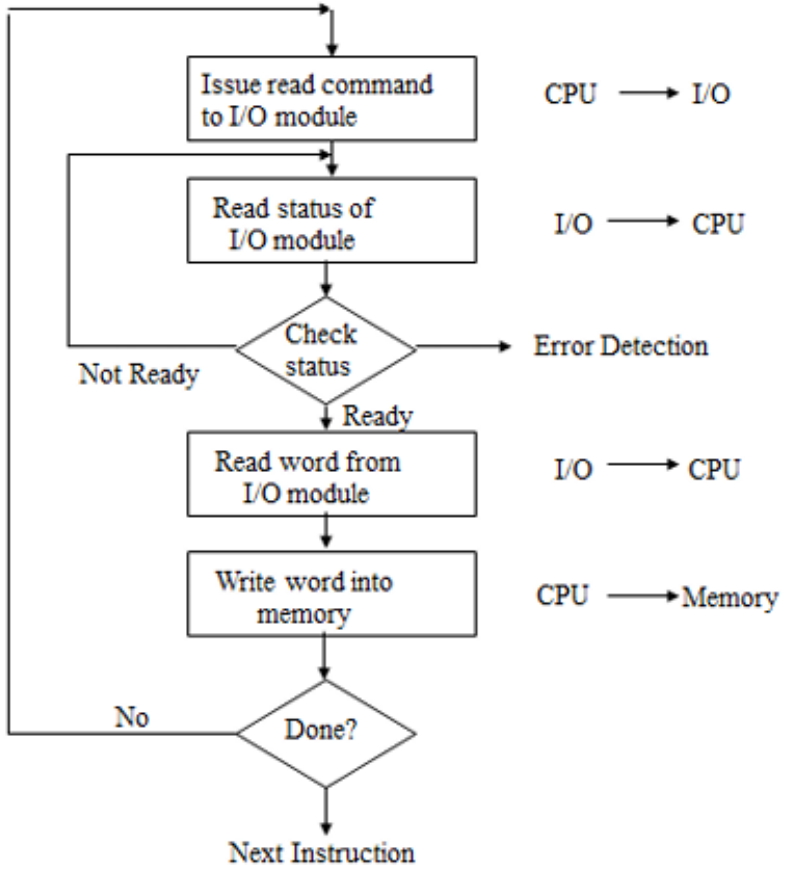



Fig. Programmed I/O operations.

# INTERRUPT DRIVEN I/O

To reduce the time spent on I/O operations for periodically checking the status of I/O device in programmed I/O, the CPU can use an interrupt – driven I/O approach. For this

- CPU issues read command to I/O module.
- I/O module gets data from peripheral while CPU does other work.
- I/O module interrupts CPU to exchange data.
- CPU requests data.
- I/O module transfers data.
- CPU issues I/O command to the module.

# INTERRUPT DRIVEN I/O

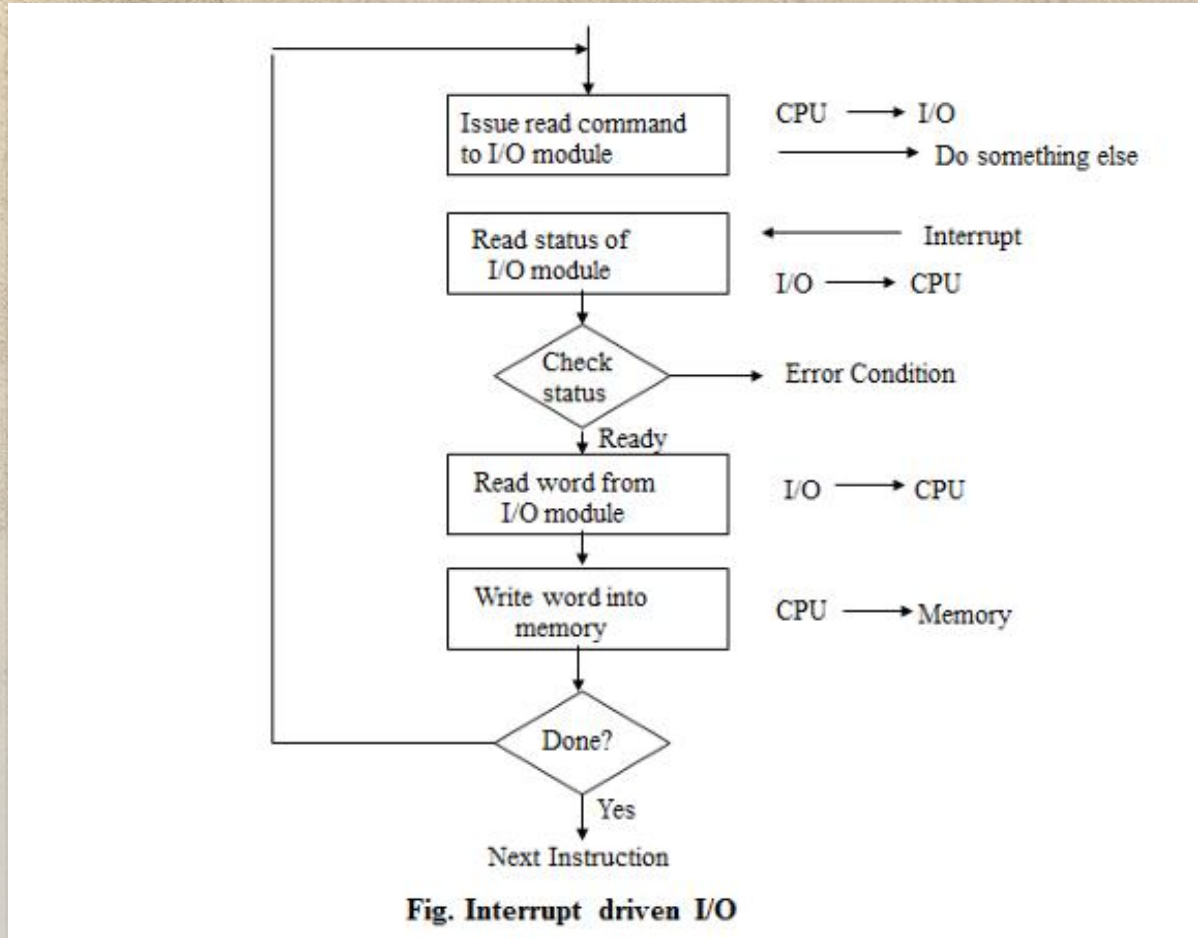



**Fig. Interrupt driven I/O**

# DIRECT MEMORY ACCESS (DMA)

DMA is used to transfer the data between memory and I/O device with less involvement of CPU. Large amounts of data can be transferred between memory and peripheral, severely impacting CPU performance. DMA involves an additional module on the system bus. DMA takes the control of system from processor.

When processor wishes to read and write a block of data, it issues a command to DMA module by sending following information:

- – Read/Write request
- – Device address
- – Starting address of memory block for data read/write
- – Amount of data to be transferred CPU carries on with other work.

DMA controller deals with transfer i.e. transfer takes place between DMA and memory without involvement of processor. DMA controller sends interrupt when finished. CPU finalizes the DMA operations and resumes. Thus processor involvement is only at the beginning and end of transfer.

# DMA MODES OF DATA TRANSFER

1. **Burst Mode-** With burst mode, data transfer takes place continuously till all the data are transferred. This mode is needed for fast devices such as magnetic disks where data transmission cannot be stopped or slowed down.

2. **Cycle Stealing Mode-** DMA controller transfers one byte of data then returns control of system buses to CPU. CPU performs an instruction, and then DMA controller transfers a data value.

3. **Transparent Mode-** DMA controller transfers data only when the CPU is performing operations that don't use system buses. Thus CPU is not idle in this case. But if CPU is using bus all the time, DMA could not take the control of system bus. This mode requires hardware to detect the state of CPU.

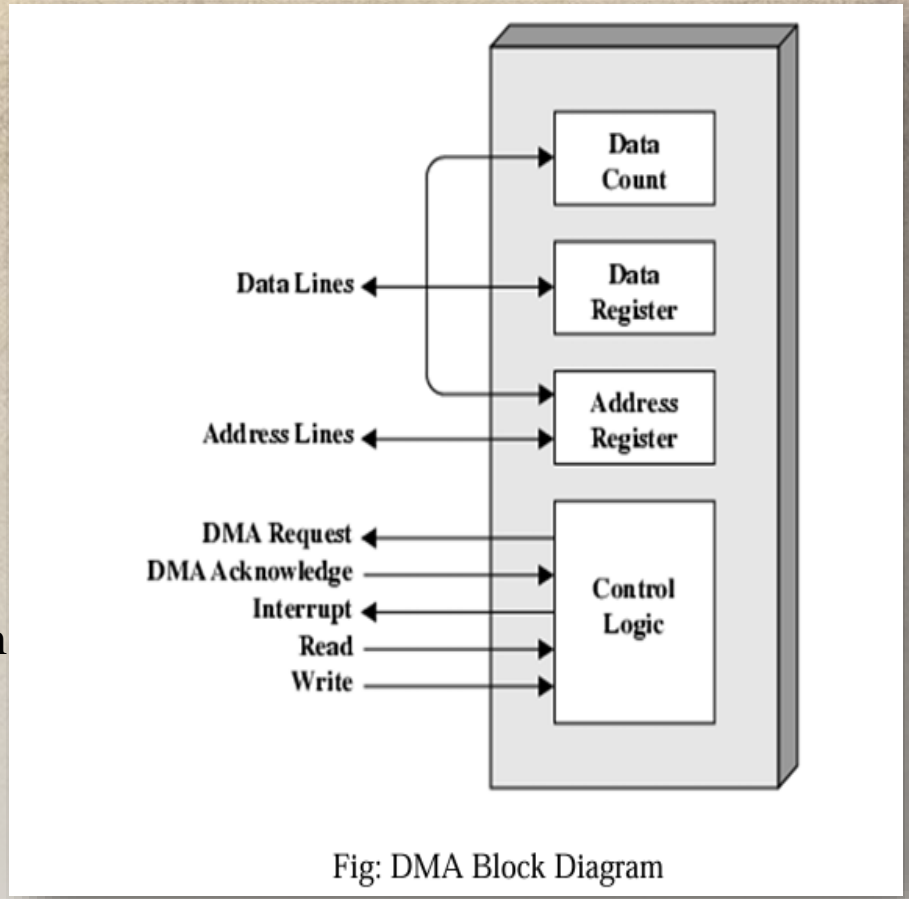


Fig: DMA Block Diagram

# I/O CHANNELS AND PROCESSORS

I/O channels are the module that behaves like a processor with specialized instruction set.

- CPU directs the processor to execute a I/O program in memory.
- I/O processor fetches an executes instruction without any involvement of CPU.
- CPU is interrupted only when I/O activities are completed.
- For I/O processors, module has its own local memory.
- With this large set of I/O devices can be controlled with minimal involvement of CPU. It controls the communication with interactive terminals.

# TYPES OF I/O CHANNELS

Two types of I/O channels:

i. **Selector-** A selector controls multiple high-speed devices but one at a time. I/O channel selects one device and data transfer takes place.

ii. **Multiplexer-** A multiplexer channel can handle multiple I/O devices at same time. For high-speed device, a block multiplexer interleaves block of data from several devices.

# THUNDERBOLT

- Thunderbolt is a high-speed input/output (I/O) technology developed by Intel that allows for fast data transfer and video streaming over a single cable.
- Thunderbolt Interface is a hardware interface used for the connection of various external peripherals for fast data transfer between a computer and the various attached peripherals.
- It supports multiple types of I/O devices, including displays, external storage devices, and network adapters.
- five generations of Thunderbolt Interface – Thunderbolt 1,2,3,4 and 5.



# INFINIBAND

**InfiniBand (IB)** is a high-speed, low-latency networking standard, designed for high-performance computing (HPC), data centers, and artificial intelligence (AI) workloads.
It is the architecture and specification for data flow between processor and intelligent I/O devices. Infiniband are intended to replace PCI(Peripheral Component Interconnect) servers increasing capacity, expandability, and flexibility.

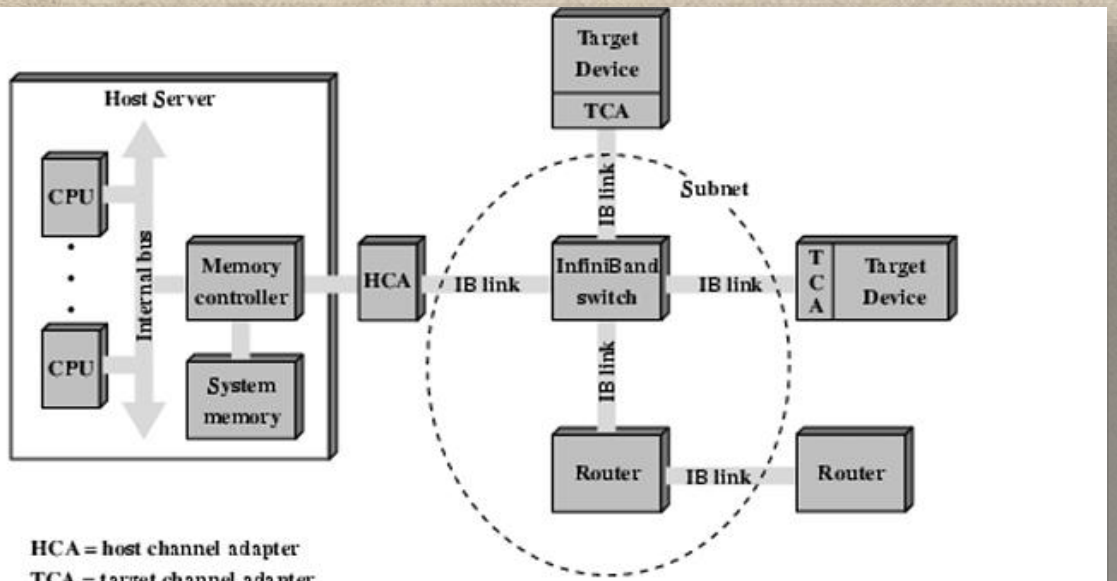



Fig: infiniband switch


THE
END